Paper D-03: Geology: A Constraint-Driven Flux Dynamics (CDFD) Perspective

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Abstract

In this paper we apply Adaptive Flux Limitation (AFL) to geology and geophysics, modelling tectonic and volcanic processes as constrained transport systems. The potential $\Phi = \Phi_{tect} + \Phi_{magma} + \Phi_{fluid}$ encodes plate motion, magmatic, and hydrothermal fluxes; constraint $C = C_{friction} + C_{crustal} + C_{viscosity}$ encodes fault friction, crustal rigidity, and magmatic viscosity. The Surface Operating Ratio $\Psi_s^* = J/C$ classifies geological states from stable inter-seismic loading through earthquake rupture and volcanic eruption.

We model the hypothesis that Gutenberg-Richter scaling, the seismic cycle, eruption styles, plate velocity-size scaling, and Venus/Earth comparisons all emerge from AFL flux–constraint dynamics.

Status: Inferred from AFL first principles; quantitative predictions are hypothesis-level.

Series tag: Dom-D03

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow–constraint language used here [3]. Part II develops the Life Number and the origins-of-life use of the same notation [4]. Part III applies AFL to biology and medicine [5]. Part IV [6], DOI `https://doi.org/10.5281/zenodo.20395901`, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [3], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \to \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

Earth's lithosphere moves at 1–10 cm/year, accumulating $\sim 10^{18}$ J/year of seismic energy at fault boundaries. This energy is released episodically in earthquakes, ranging from magnitude 1 (microseismic) to magnitude 9.5 (1960 Valdivia event, which released $\sim 10^{23}$ J). The Gutenberg-Richter law — the power-law frequency-magnitude relationship observed in every seismic catalogue worldwide — is one of geology's most robust empirical observations, yet its mechanistic origin has remained debated.

Volcanic systems provide a parallel AFL process: magma generated by partial melting of the mantle must overcome crustal constraint to erupt. The 1991 Pinatubo eruption injected $\sim$10 km^3 of material and temporarily cooled global average temperature by 0.5°C — an AFL constraint failure of planetary significance.

AFL provides a unified framework for both seismic and volcanic systems: both are instances of potential accumulation against constraint, with episodic failure when $\Psi_s^* > 1$. The AFL geological framework connects to the Part IV Earth series: volcanic outgassing drives atmospheric Φ (Paper 02); tectonic cycling controls long-term carbon sequestration and climate (Paper 10); earthquake cascades resemble the systemic crisis propagation model of Paper 33.

3 Core Hypothesis

Geological systems are AFL flux–constraint systems in which tectonic and magmatic potential Φ accumulates against crustal, frictional, and viscous constraint C: earthquakes are AFL phase transitions when $\Psi_{s,seismic} > 1$ and fault friction is overcome; volcanoes erupt when $\Psi_{s,magma} > 1$ and crustal strength is exceeded; the Gutenberg-Richter law emerges from AFL constraint heterogeneity across the crust.

AFL geological hypothesis: the seismic cycle — inter-seismic loading, critical state, rupture, post-seismic relaxation — is an AFL limit cycle:

$$\frac{d\sigma}{dt} = \dot{\sigma}_{loading} - \gamma_{rupture}\sigma \cdot \mathbb{1}[\sigma > \sigma_{cr}], \tag{1}$$

where σ is fault stress (Φ), $\dot{\sigma}_{loading}$ is the tectonic loading rate (S), and $\gamma_{rupture}$ is the rupture propagation rate. The Gutenberg-Richter exponent $b \approx 1$ emerges from AFL constraint heterogeneity: if $C_{friction}$ is exponentially distributed across the fault system, then earthquake magnitudes (proportional to $\log(A_{rupture})$) follow a power law. **Status: Hypothesis; Gutenberg-Richter derivation requires AFL heterogeneity analysis.**

4 Geological Potential Fields

We define the generalised geological potential:

$$\Phi = \Phi_{tect} + \Phi_{magma} + \Phi_{fluid} \tag{2}$$

where:

- Φ_{tect} — tectonic stress potential (accumulated elastic strain energy at fault systems; primary driver of seismicity; scales with plate convergence/divergence rate and fault locking degree);
- Φ_{magma} — magmatic pressure potential (melt fraction $\times$ density contrast; drives volcanic systems; accumulates in magma reservoirs over decades to centuries; released explosively or effusively when crustal constraint is overcome);
- Φ_{fluid} — hydrothermal fluid potential (pore pressure in fault zones; reduces effective normal stress; key trigger for induced seismicity from wastewater injection).

5 Geological Constraint Axes

$$C = C_{friction} + C_{crustal} + C_{viscosity} \tag{3}$$

Table 1: Geological constraint axes and AFL mechanisms.

Constraint axis	Geological substrate	AFL role
$C_{friction}$	Fault zone friction (Byerlee's law)	Primary seismic constraint; overcomes at rupture
$C_{crustal}$	Crustal rigidity and yield strength	Volcanic pathway constraint; fails at eruption
$C_{viscosity}$	Magma viscosity (silica, temperature)	Eruption style control; high viscosity = explosive
C_{pore}	Fluid pore pressure	Reduces effective friction; induced seismicity driver
$C_{overburden}$	Lithostatic pressure	Depth-dependent constraint on all geological fluxes

5.1 Adaptive Surface Dynamics: Memory and Responsiveness

The operating ratio Ψ_s^* is mediated by adaptive surface components:

$$\Psi_s^* = (\Phi/C) \cdot S \cdot M_s \tag{4}$$

where S (Surface Responsiveness) and M_s (Surface Memory) evolve as:

$$\frac{dS}{dt} = \kappa_s(\Psi_s^* - S) \tag{5}$$

$$\frac{dM_s}{dt} = \Phi S - d_m M_s \tag{6}$$

These components represent the homeostatic capacity of the system to regulate its own operating ratio through memory and responsiveness.

6 Flux Law

Tectonic and volcanic throughput:

$$J_{geological} = -\frac{1}{C}\nabla\Phi \tag{7}$$

where:

- For seismic systems: $J_{seismic}$ = fault slip rate; $\nabla\Phi$ = stress gradient along fault; $C = C_{friction}$ (Byerlee friction coefficient $\times$ normal stress);
- For volcanic systems: $J_{volcanic}$ = magma ascent rate (m/s); $\nabla\Phi$ = magma pressure gradient (overpressure relative to lithostatic); $C = C_{viscosity} + C_{crustal}$;
- For hydrothermal systems: J_{fluid} = fluid flux (Darcy's law); $\nabla\Phi$ = pore pressure gradient; $C = \text{permeability}^{-1}$ (inverse permeability).

7 Conservation Dynamics

Geological potential evolves as:

$$\frac{\partial\Phi}{\partial t} = \nabla \cdot \left(\frac{1}{C}\nabla\Phi\right) + S \tag{8}$$

where S represents:

- **Tectonic loading**: $S_{tect} = \dot{\sigma}_{plate}$ (far-field plate velocity $\times$ elastic modulus / fault locked length; steady background loading);
- **Magma generation**: S_{magma} from mantle partial melting (proportional to mantle temperature anomaly and volatile content);
- **Glacial unloading**: $S_{glacio} > 0$ as ice sheets melt (reduces $C_{overburden}$, triggering seismicity and volcanism at high latitudes — observed in Iceland and Svalbard).

8 Adaptive Constraint Dynamics

$$\frac{dC}{dt} = \alpha_{geo}|J_{geological}| - \beta_{geo}C \tag{9}$$

where:

- α_{geo}: constraint stress from geological throughput (fault slip raises temperature and gouge thickness, increasing $C_{friction}$ transiently via velocity-strengthening behaviour on some faults; magma flow erodes conduit walls, reducing $C_{viscosity}$);
- β_{geo}: geological recovery rate (fault healing between earthquakes; magma cooling in conduit; post-seismic viscoelastic relaxation restoring $C_{crustal}$).

8.1 Adaptive Surface Dynamics: Memory and Responsiveness

The operating ratio Ψ_s^* is mediated by adaptive surface components:

$$\Psi_s^* = (\Phi/C) \cdot S \cdot M_s \tag{10}$$

where S (Surface Responsiveness) and M_s (Surface Memory) evolve as:

$$\frac{dS}{dt} = \kappa_s(\Psi_s^* - S) \tag{11}$$

$$\frac{dM_s}{dt} = \Phi S - d_m M_s \tag{12}$$

These components represent the homeostatic capacity of the system to regulate its own operating ratio through memory and responsiveness.

9 Flux–Constraint Number

$$\Psi_{s,geological} = \frac{J_{geological}}{C_{friction} + C_{crustal} + C_{viscosity}} \tag{13}$$

9.1 Inter-seismic Regime ($\Psi_s^* < 0.8$)

Fault is locked; stress accumulates but does not exceed friction. Geodetic monitoring shows surface deformation consistent with elastic strain accumulation. AFL $dC/dt > 0$: fault strengthens (velocity-strengthening behaviour). Typical duration: decades to centuries depending on plate velocity and fault locking coefficient.

9.2 Critical Seismic Regime ($\Psi_s^* \approx 1$)

Fault stress approaches friction threshold. Microseismicity increases (small AFL Ψ_s^* excursions at local heterogeneities); Coulomb stress transfer from nearby earthquakes can trigger the main shock. AFL predicts critical slowing down: variance of microseismicity rate increases, and autocorrelation length increases — consistent with seismic early-warning research.

9.3 Rupture/Eruption Regime ($\Psi_s^* > 1.2$)

Fault friction or crustal strength is overcome. Earthquake rupture propagates at 2–3 km/s; energy release is $E_{seismic} = \int_A (\sigma - C_{friction})\, dA$. AFL cascade: stress drop from rupture raises Ψ_s^* on adjacent fault segments (Coulomb stress transfer), triggering aftershocks.

10 Gutenberg-Richter Law as AFL Constraint Heterogeneity

The Gutenberg-Richter law:

$$\log_{10} N(M) = a - bM, \quad b \approx 1.0, \tag{14}$$

relates earthquake magnitude M to annual frequency $N(M)$. AFL derives this from constraint heterogeneity: if $C_{friction}$ is heterogeneously distributed across the fault system with an exponential distribution $P(C) \propto e^{-C/\bar{C}}$, then the probability that an AFL excursion ($\Psi_s^* > 1$ locally) grows to magnitude M is:

$$P(A_{rupture} > A) \propto A^{-b}, \quad b = \left.\frac{\partial \ln P(C)}{\partial C}\right|_{C=C_{mean}}. \tag{15}$$

AFL predicts $b = 1$ when constraint is exponentially distributed; $b < 1$ when constraint is heterogeneous (heterogeneous crust, distributed active faults); $b > 1$ when constraint is homogeneous (single locked fault with uniform friction). Observed b variations (0.7–1.5) reflect AFL constraint heterogeneity variations across different tectonic settings.

11 Seismic Cycle as AFL Limit Cycle

The seismic cycle is an AFL limit cycle:

$$\frac{d\Phi_{stress}}{dt} = \dot{\sigma}_{tect} - J_{seismic} \cdot \mathbb{1}[\Psi_s^* > \Psi_{s,cr}] \tag{16}$$

$$\frac{dC_{friction}}{dt} = \alpha_{heal} - \beta_{damage} C_{friction} \cdot \mathbb{1}[\Psi_s^* > \Psi_{s,cr}] \tag{17}$$

Cycle phases:

1. **Inter-seismic**: Φ rises at rate $\dot{\sigma}_{tect}$; $C_{friction}$ heals slowly; $\Psi_s^* < 1$ throughout;
2. **Co-seismic**: $\Psi_s^* > 1$ threshold crossed; rupture propagates; $\Phi_{stress} \to 0$ and $C_{friction}$ drops to dynamic friction value;
3. **Post-seismic**: afterslip and viscoelastic relaxation; $\Psi_s^* \to 0$ as stress drops; $C_{friction}$ begins healing;
4. **Return to inter-seismic**: $C_{friction}$ heals back to full inter-seismic value over $\tau_{heal} = 1/\beta_{heal}$ (fault recurrence interval).

The recurrence interval $T_{rec} = (\sigma_{cr} - \sigma_0)/\dot{\sigma}_{tect}$ is the AFL prediction for earthquake return periods on a specific fault, derivable from geodetic plate velocity measurements and seismic stress-drop estimates.

12 Volcanic Systems as AFL Runaway Events

Volcanic eruptions are AFL runaway events where $\Psi_{s,magma} > 1$:

$$\Psi_{s,magma} = \frac{\Phi_{magma-pressure}}{C_{crustal} + C_{viscosity}} > 1 \quad \Rightarrow \quad \text{eruption.} \tag{18}$$

Table 2: Eruption styles as AFL $\Psi_{s,magma}$ and constraint decomposition.

Eruption style	$\Psi_{s,magma}$	$C_{viscosity}$	$C_{crustal}$	Example
Hawaiian effusive	1.05–1.2	Low (basalt)	Low (thin crust)	Kilauea
Strombolian	1.2–1.5	Moderate	Moderate	Stromboli
Vulcanian	1.5–2.0	High	Moderate	Soufriere Hills
Plinian (explosive)	>2.0	Very high (rhyolite)	High (thick crust)	Pinatubo, St. Helens

Magma viscosity $C_{viscosity}$ is the key AFL constraint determining eruption explosivity:

$$C_{viscosity} \propto \exp\left(\frac{E_{viscosity}}{RT}\right) \cdot (1 + [\mathrm{SiO_2}]/SiO_{2,ref}), \tag{19}$$

where $E_{viscosity}$ is the viscous activation energy and $[\mathrm{SiO_2}]$ is the silica content. Basaltic magmas (low SiO_2) have $C_{viscosity} \ll 1$ (low constraint, effusive eruption); rhyolitic magmas (high SiO_2) have $C_{viscosity} \gg 1$ (high constraint, explosive eruption when overpressure exceeds $C_{viscosity}$).

13 Plate Tectonics as Planetary-Scale AFL

Plate tectonics represents the Earth's $\Psi_s^* \approx 1$ regime for mantle heat flux:

$$\Psi_{s,tectonic} = \frac{J_{mantle-convection}}{C_{lithospheric-rigidity}} \approx 1. \tag{20}$$

AFL planetary comparison:

- **Earth**: $\Psi_{s,tectonic} \approx 1$ (plate tectonics; efficient mantle heat transport; carbon cycle maintained; surface water reduces $C_{lithospheric}$ via serpentinisation and hydrous weakening);
- **Venus**: $\Psi_{s,tectonic} < 0.5$ (stagnant lid; $C_{lithospheric} \gg J_{mantle}$; no surface water to weaken crust; heat escapes via episodic catastrophic resurfacing every 300–500 Myr rather than continuous plate tectonics);
- **Mars**: $\Psi_{s,tectonic} \to 0$ (dead planet; mantle cooled; all constraint; no flux; no plate tectonics; magnetic field dead since 4 Gyr ago).

AFL predicts that water is essential for maintaining $\Psi_{s,tectonic} \approx 1$ on terrestrial planets: water reduces $C_{lithospheric}$ via serpentinisation, enabling plate tectonics, which then maintains the carbon cycle and surface water — a positive feedback maintaining habitability.

14 Induced Seismicity as AFL Constraint Reduction

Wastewater injection from hydraulic fracturing raises pore pressure Φ_{fluid}, reducing effective normal stress and thus $C_{friction}$:

$$C_{friction,eff} = C_{friction,dry} - \Phi_{pore-pressure}, \tag{21}$$

driving $\Psi_{s,seismic} \to 1$ on critically stressed faults. AFL predicts induced seismicity magnitude from injection volume and formation permeability: high permeability allows Φ_{fluid} to reach distant pre-existing faults at higher Ψ_s^*, enabling larger earthquakes. The 2011 Oklahoma induced earthquake sequence ($M > 5.0$) is the AFL consequence of $\Psi_{s,seismic} > 1$ from regional disposal well injection.

15 Spatial Self-Organisation of Seismic Systems

Fault networks organise into hierarchical systems:

- **Transform faults**: low-C pathways where plates slide past each other; AFL routing along minimum-resistance path;
- **Subduction zones**: high-Φ convergence zones; greatest seismic energy release;
- **Aftershock sequences**: AFL cascade from main shock reducing C on adjacent segments; Omori law ($n(t) \propto t^{-p}$, $p \approx 1$) is the AFL relaxation rate.

The Omori aftershock decay law is the AFL post-rupture constraint recovery:

$$n(t) = \frac{K}{(t+c)^p}, \quad p = \frac{\beta_{heal}}{\alpha_{damage}} \approx 1. \tag{22}$$

16 Predictions

1. **Gutenberg-Richter b-value**: AFL predicts $b \propto 1/\mathrm{Var}(\ln C_{friction})$; heterogeneous fault systems have lower b (more large earthquakes); this is testable from fault geometry mapping and laboratory friction experiments;
2. **Seismic cycle timing**: recurrence interval $T_{rec} \propto (\sigma_{cr} - \sigma_0)/\dot{\sigma}_{tect}$; AFL predicts earthquake timing from geodetic plate velocity data;
3. **Eruption precursors**: $d\Psi_{s,magma}/dt > 0$ for 6–18 months before large eruptions; AFL predicts increasing ground inflation (rising Φ_{magma}), increasing seismicity rate ($J_{magmatic}$), and increasing harmonic tremor as precursors;
4. **Induced seismicity**: test a candidate relation between injected volume, mapped fault constraint, and maximum observed moment:
$$\log_{10} M_0 \propto \log_{10}\left(V_{inject}/\sqrt{C_{friction}}\right).$$
The relation must be compared with established induced-seismicity models;
5. **Planetary habitability**: AFL predicts that terrestrial planets with surface water (lower $C_{lithospheric}$) maintain $\Psi_{s,tectonic} \approx 1$ and are more likely to develop life than arid planets with $\Psi_{s,tectonic} \ll 1$ (stagnant lid).

Status: Predictions 1–3 supported by seismological literature; predictions 4–5 are AFL-derived hypotheses.

17 Computational Interpretation

The companion script `supplementary_dom_03.py` implements AFL geological dynamics:

- seismic cycle: $\Phi_{stress}(t)$ and $C_{friction}(t)$ ODE integration; rupture at $\Psi_{s,cr}$; Omori aftershock sequence generated post-rupture;
- Gutenberg-Richter: heterogeneous $C_{friction}$ distribution; b-value extracted from simulated earthquake catalogue;
- volcanic: $\Psi_{s,magma}(t)$ under magma supply; eruption onset; style classified from $C_{viscosity}$;
- induced seismicity: pore pressure diffusion on fault network; $\Psi_{s,seismic}$ evolution with injection;
- outputs: seismic cycle $\Psi_s^*(t)$, G-R frequency-magnitude plot, eruption timeline.

18 Discussion

AFL provides a unified framework for geology: earthquakes are AFL phase transitions; the seismic cycle is an AFL limit cycle; the Gutenberg-Richter law is AFL constraint heterogeneity; volcanic eruption style is determined by $\Psi_{s,magma}$ and the constraint decomposition ($C_{viscosity}$ vs $C_{crustal}$); plate tectonics is the planetary-scale AFL $\Psi_s^* \approx 1$ regime for mantle heat transport.

The AFL geological framework connects to the Part IV Earth series: tectonic outgassing drives the carbon cycle (Paper 10); volcanic aerosols modulate climate via AFL atmospheric forcing (Paper 02); earthquake cascades resemble the systemic crisis propagation model (Paper 33). The water-tectonic-habitability connection identified by AFL also links to the sustainability synthesis (Paper 10).

19 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

19.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [7], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub $\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

20 Major Universal Upgrade: Mechanism, Scale, and Tests

20.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Geology, the proposed drive is tectonic stress, heat, fluids, sediment, and chemical gradients. The active limitation is rock strength, friction, pressure, permeability, and geometry. Responsiveness is carried by deformation, fracture, flow, phase change, and chemical alteration, while retained state is represented by stratigraphy, faults, fabric, thermal state, and prior deformation. The outcome to explain is landforms, hazards, material transport, and geological transition.

Table 3: Operational CDFD/AFL map for Paper D-03.

Term	**Paper-specific observable meaning**	**Measurement question**
Φ	tectonic stress, heat, fluids, sediment, and chemical gradients	What rate, load, gradient, or demand enters the focal system?
C	rock strength, friction, pressure, permeability, and geometry	Which measured bottleneck limits transfer, function, or recovery?
S	deformation, fracture, flow, phase change, and chemical alteration	Which process changes effective capacity or reroutes the load?
M_s	stratigraphy, faults, fabric, thermal state, and prior deformation	Which retained state makes equal present loads produce different futures?
Y	landforms, hazards, material transport, and geological transition	Which outcome is predicted out of sample, with uncertainty?

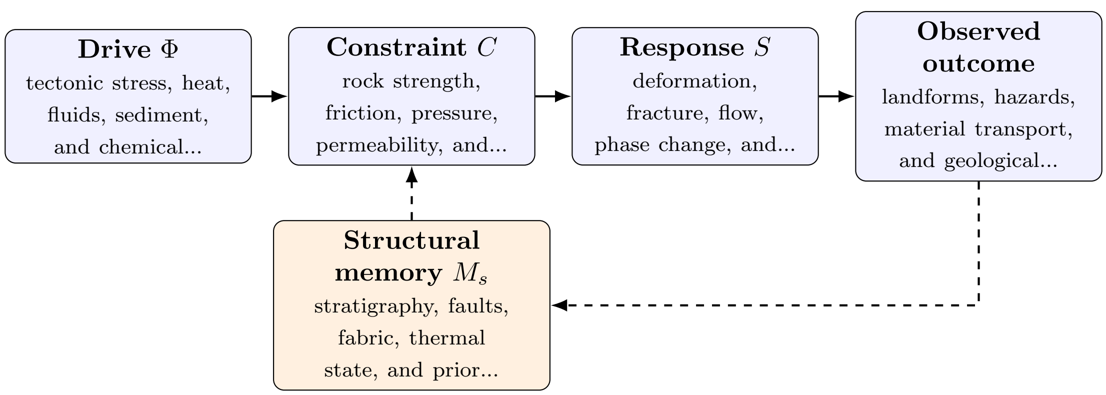

Figure 1: Paper D-03 observable CDFD/AFL architecture for Geology. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

20.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \tag{23}$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \tag{24}$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \tag{25}$$

$$\Psi_s(t) = \frac{\widehat{\Phi}(t)}{\epsilon + \widehat{C}(t)}\,\widehat{S}(t)\,[1 + \omega\widehat{M_s}(t)]. \tag{26}$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in tectonic stress, heat, fluids, sediment, and chemical gradients arrives faster than deformation, fracture, flow, phase change, and chemical alteration can compensate. This raises or redistributes rock strength, friction, pressure, permeability, and geometry. If the load persists, the retained state encoded by stratigraphy, faults, fabric, thermal state, and prior deformation changes the next response, so a later event of equal magnitude can produce a different trajectory in landforms, hazards, material transport, and geological transition. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

20.3 Cross-Scale Universality Without Mechanistic Erasure

For the domain applications family, the universal comparison is a disciplined translation test across systems with very different material mechanisms. A valid mapping preserves the baseline science of the domain, declares the scale of every variable, and earns its place through a discriminating prediction rather than resemblance alone. [2, 9, 8, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [3]; Part II asks when maintained throughput supports persistent organized states [4]; Part III develops adaptive limitation and biological memory [5]; and Part IV [6] tests how much of that architecture survives translation. The CDFD Runtime [7] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is seconds to billions of years; grains to planets. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

20.4 Evidence Ladder and Discriminating Tests

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase

Table 4: Evidence ladder for Paper D-03. Each rung can fail independently.

Test	**Design**	**Discriminating result**
Proxy audit	Measure field mapping, geodesy, seismology, petrology, geochemistry, and stratigraphic records and preregister how each record maps to Φ, C, S, M_s, and Y.	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe a stress or fluid-pressure perturbation across inherited structures while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: explicit memory and topology do not improve geological interpretation or prediction.

estimates β and μ, separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

20.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from field mapping, geodesy, seismology, petrology, geochemistry, and stratigraphic records and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of landforms, hazards, material transport, and geological transition. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

21 Conclusion

Geological systems are AFL flux–constraint systems operating at planetary scales. Earthquakes, volcanic eruptions, plate tectonics, and induced seismicity all emerge from the universal AFL equation with domain-appropriate (Φ, C) interpretation. The Gutenberg-Richter law, seismic cycle periodicity, and eruption style are AFL predictions derivable from constraint heterogeneity and the ratio $\Psi_s^* = J/C$.

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